

Fanny Traore

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EDUCATION

Carleton University – B.Eng. in Computer Systems Engineering (Co-op) Expected 2029
Completed Coursework: Object-Oriented Software Development (Java, UML, JUnit), Algorithms & Data Structures (Python), Foundations of Imperative Programming (C, Linux), Computer Organization & Architecture, Signals and Systems, Intro to Digital Systems, Data Management
In Progress (Fall 2026): Operating Systems, Intro to Real-Time Systems (embedded C on a TI MSP432 microcontroller)

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, HTML/CSS, MATLAB, Verilog, Bash/Shell, PowerShell
Testing & Tools: pytest, Jest, JUnit, Python unittest, GitHub Actions (CI), Git, GitHub (pull request reviews), Jira, VS Code, Vivado, Figma
Frameworks: Node.js, Electron, npm, Chrome Extensions (Manifest v3), REST APIs
Hardware & Lab: Arduino, TI MSP432, FPGA boards (PYNQ-Z2, Zybo Z7), UART serial, oscilloscope, logic analyzer, multimeter, function generator, KiCad
Operating Systems: Linux (Ubuntu), Windows **Spoken Languages:** English, French (bilingual)

PROJECTS

Mochadoro: Desktop App + Arduino Hardware | *Electron, Node.js, JavaScript, HTML/CSS, C++* Dec 2025 – Sept 2026

- Designed and coded the entire user interface of a desktop productivity app in HTML, CSS and JavaScript, planning the layout in Figma and drawing all visual assets.
- Achieved sub-50 ms bidirectional sync between the app and Arduino hardware through state-machine session tracking over UART serial (9600 baud).
- Programmed C++ firmware for RGB status LEDs, a 7-segment cycle counter and a piezo buzzer.
- Found the root cause of a dark display segment by checking the wiring against the datasheet pinout and the firmware pin IDs: the display had been inserted backwards.

ScamFree: Phishing Detection (Hackathon, team of 2) | *JavaScript, Python, GitHub Actions* Jan 2026 – Sept 2026

- Built the Manifest v3 Chrome extension that shows non-blocking phishing alerts in Gmail using the team's Node.js REST API.
- Wrote a Python data-validation module and 10 of its 11 pytest tests, set up a 7-test Jest suite with a 95% accuracy quality gate, and ran both in GitHub Actions CI on every push and pull request.
- Fixed a data-loader defect caught by the tests, logged it as a GitHub Issue, and protected the main branch behind passing checks after a CI run showed the fix had never reached it.
- Linked 8 *shall* requirements to their tests in a requirements traceability matrix.

PYNQ-Z2 Morse Code Decoder | *Verilog, Vivado, FSM Design* Sept 2026 – Present

- Built a jitter-tolerant Morse code decoder on the Zynq-7020 FPGA with a metastability synchronizer, a 125 MHz/17-bit clock divider and a 4-state FSM in Verilog.
- Isolated and eliminated FSM chatter found during hands-on testing by routing state transitions to LEDs and adding multi-cycle pulse conditioning.

EXPERIENCE

FPGA Summer Club – Team Leader, Carleton University May 2026 – Aug 2026

- Led a three-person team building a real-time video filter on a Digilent Zybo Z7 board, authoring the XDC pin-constraint file that mapped the design to board I/O.

Carleton Planetary Robotics (CPRT) – Software Team Member, Carleton University Sept 2025 – Dec 2025

- Evaluated rover vibration profiles and proposed elastomer mounts and aluminum bracing for sensor stacks and electrical enclosures; used Jira for team task tracking.

Bridgehead – Barista 2026 – Present

- Serve customers face to face in a busy coffee shop, taking and preparing orders.

Popeyes Louisiana Kitchen – Crew Member and Cashier 2022 – 2024

- Served customers face to face as a cashier in a fast-paced fast-food restaurant, taking orders and payments.